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Index

1. Introduction	2
2. Flowchart	3
3. Project Overview	4
4. Configuration Steps	5
5. Testing & Verification	8
6. Conclusion	12

Design and Simulation of Smart Home Security Ecosystem using Cisco Packet Tracer

Introduction

With the rapid advancement of the Internet of Things (IoT), smart home automation and remote security systems have become essential components of modern living. This project focuses on designing and implementing a secure, scalable, and automated **Smart Home Security Ecosystem** using **Cisco Packet Tracer**.

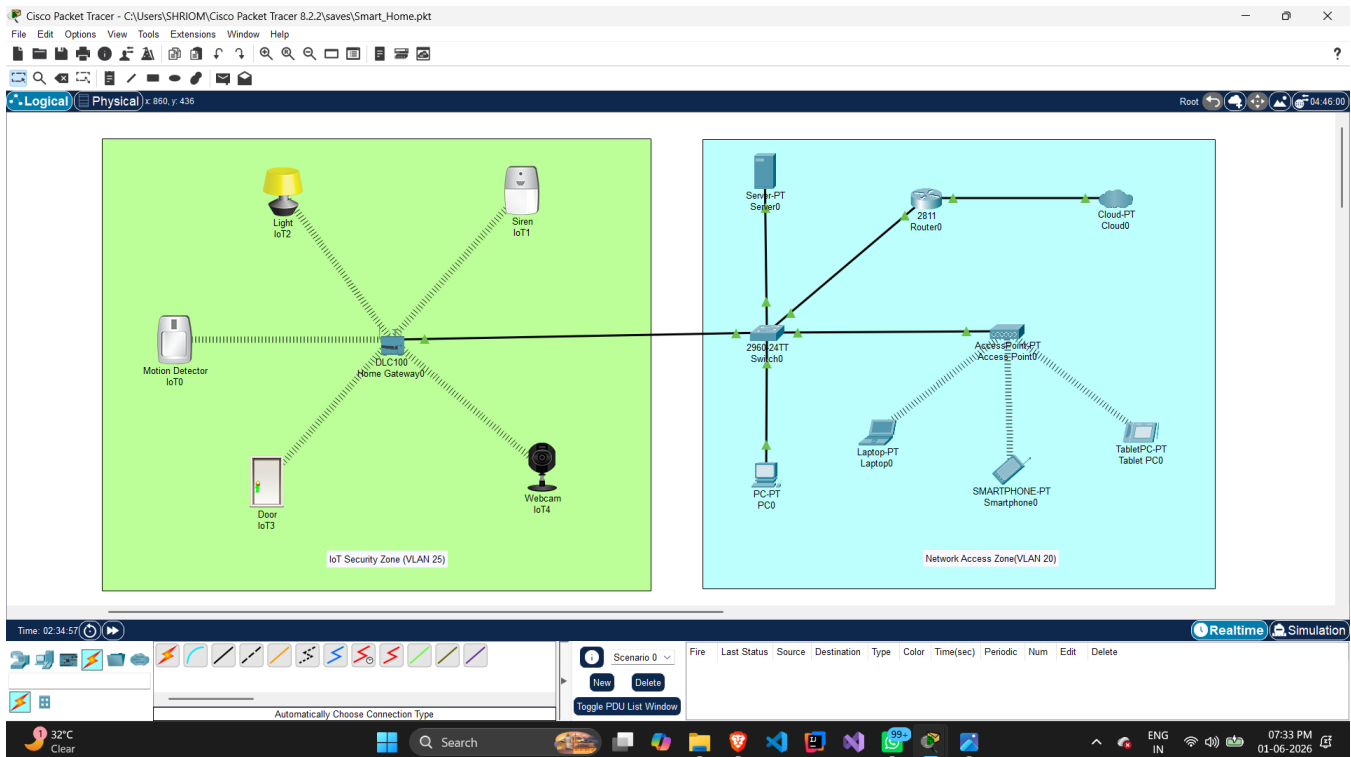
The primary objective of this system is to establish an intelligent environment that can autonomously detect security breaches, trigger immediate local defensive actions, and alert the user remotely in real-time.

The system integrates multiple IoT devices such as motion detectors, smart lights, sirens, webcams, and a home gateway to provide automated security monitoring. When motion is detected, the system automatically activates security devices, including alarms and lighting, and generates alert notifications to inform the user about potential intrusions.

To improve network security and organization, the network is divided into two Virtual Local Area Networks (VLANs). VLAN 25 is for IoT security devices and VLAN 20 is for user and network access devices such as PCs, laptops, smartphones, tablets, servers, routers, and access points. This VLAN-based segmentation improves security, reduces unnecessary traffic and allows for better network management.

Flowchart Of Smart Home Security Ecosystem





1. Project Overview

The architecture is strategically divided into two distinct logical networks through **VLAN Segmentation** to ensure maximum data privacy, network efficiency, and security:

- **IoT Security Zone (VLAN 25):** Dedicated entirely to critical smart home appliances, including a Motion Detector, Smart Light, Siren, Automated Door, Webcam, and the central Home Gateway.
- **Network Access Zone (VLAN 20):** Dedicated to user end-devices such as the Main Server, Router, Switch, Wireless Access Point, Laptops, Smartphones, and Tablets used for monitoring.

Core Working Principle

The operational flow of the system relies on automated event-driven triggers:

1. **Detection:** The system constantly monitors the environment via the Motion Detector.
2. **Local Threat Mitigation:** Once motion is detected, the system immediately executes local protocols—activating the Siren to sound an audible alarm and

turning on the Smart Light. Concurrently, the Webcam is activated to capture real-time footage.

3. **Remote Notification:** To keep the homeowner informed, the central system routes an urgent alert through the network, sending an email notification via SMTP. The user successfully receives the alert on their smartphone or tablet, allowing them to securely access the IoT Monitor interface and view live device statuses.

By utilizing **Inter-VLAN Routing** via a router-on-a-stick configuration, this project successfully bridges the gap between secure IoT automation and cross-network remote management, mimicking an industry-standard enterprise deployment.

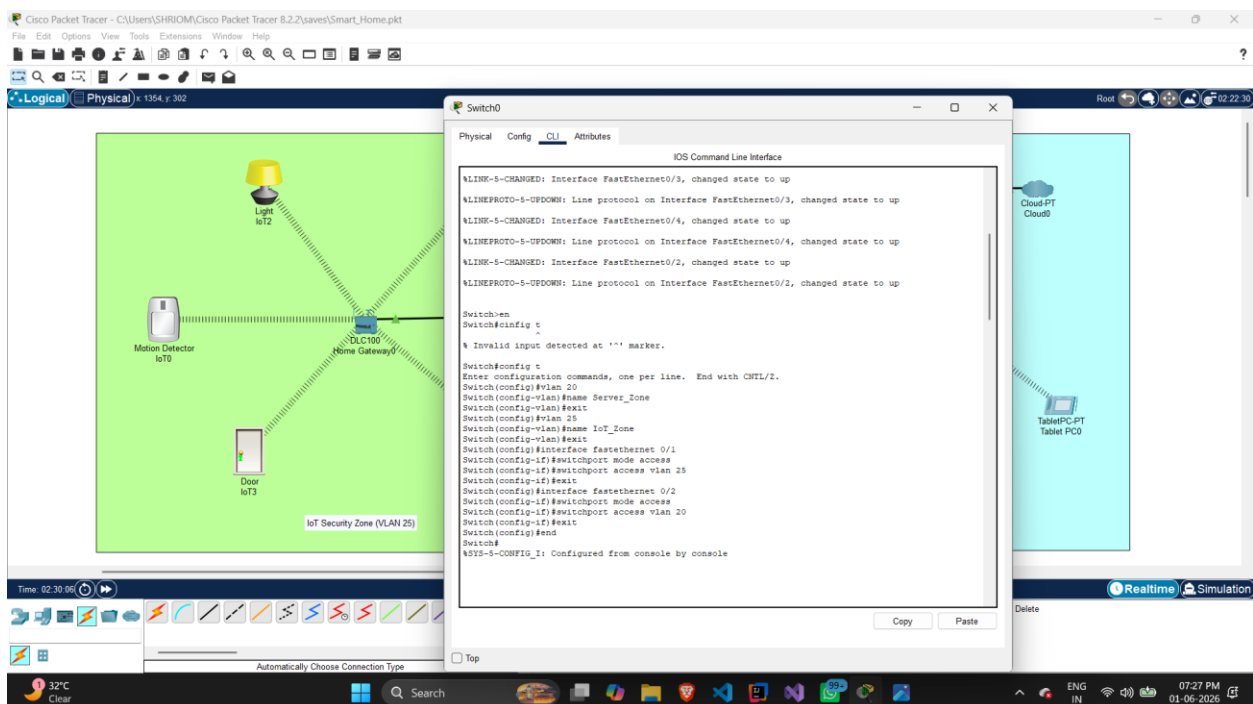
2. Configuration Steps

The implementation of the Smart Home Security Ecosystem is completed in five systematic stages to ensure stable deployment and proper Inter-VLAN routing.

Step 2.1: VLAN and Trunking Configuration on Switch

First, create the separate broadcast domains (VLANs) on **Switch0** and enable trunking on the uplink connection to the router:

- Access the Switch CLI and enter global configuration mode.
- Create VLAN 20 and VLAN 25:

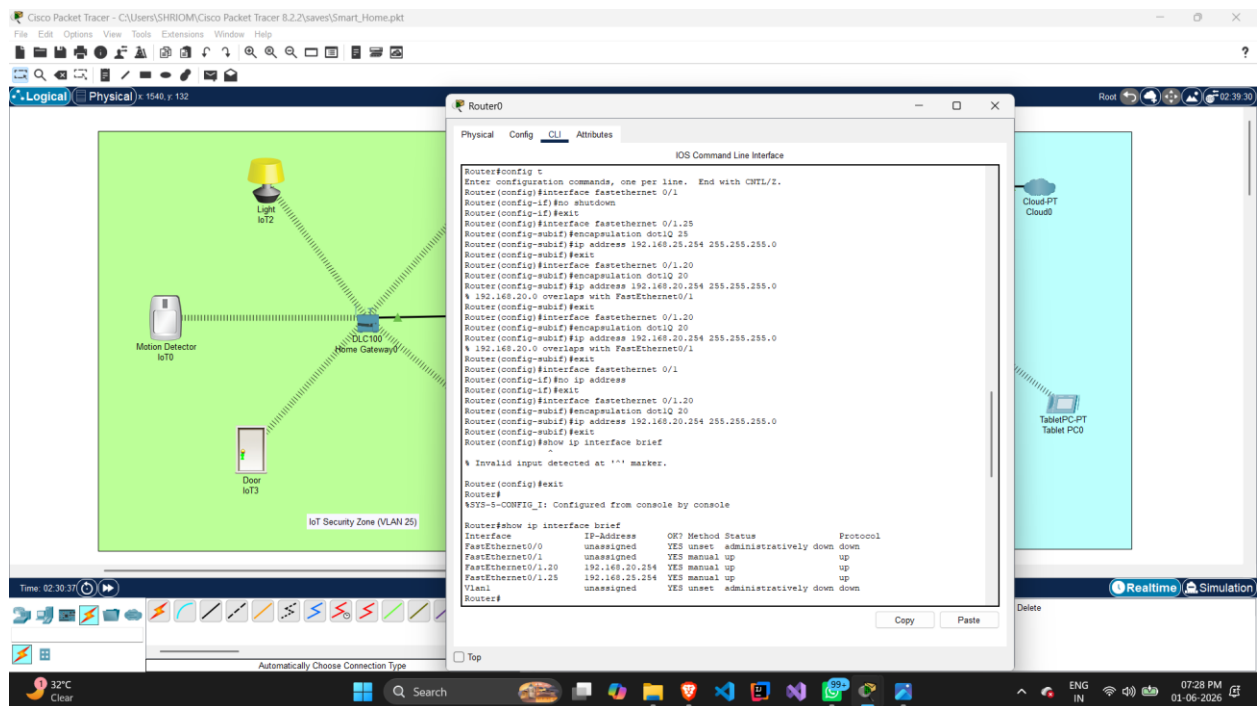


- Assign the switch ports connected to **Server**, **AccessPoint**, and end-user devices to VLAN 20.
- Assign the switch port connected to **Home Gateway** to VLAN 25.
- Configure the interface connecting to **Router** as a **Trunk Port** to allow multi-VLAN traffic flow.

Step 2.2: Inter-VLAN Routing Configuration on Router (Router-on-a-Stick)

Configure sub-interfaces on **Router**'s physical port to serve as the default gateways for both network zones:

- Initialize the physical interface and clear any base IP setups.
- Configure the sub-interface for the User Zone (**VLAN 20**):
- Configure the sub-interface for the IoT Gateways Zone (**VLAN 25**):



Step 2.3: Wireless Access Point and End-User Setup

Establish wireless data pathways for user appliances in the network access zone:

- Configure **AccessPoint** with a standard SSID profile (e.g., **Default**) running basic wireless broadcasts.
- Access the wireless interface profiles on the **Tablet**, **Laptop**, and **Smartphone**.

- Point their active wireless adaptors toward the designated SSID to establish link handshakes.
- Toggle IP profiles to **DHCP** to ensure they receive working IP coordinates within the **192.168.20.X** scope automatically.

Step 2.4: IoT Security Zone Deployment & Registration

Set up the localized smart cluster to interact seamlessly with the central home manager:

- Verify that **Home Gateway**'s Internet uplink interface is running on DHCP to dynamically fetch an active outside reference IP (e.g., **192.168.20.1**) from the distribution path.
- Power up the environmental components: **Motion Detector, Smart Light, Siren, Automated Door, and Webcam.**
- Navigate to the **Config** window of each individual IoT device.
- Set the **IoT Server** setting to **Home Gateway** to route all tracking metrics directly through the localized smart controller.

Step 2.5: Server Configuration and Automated Rules Setup

Establish centralized application services for mail distribution and conditional logic mapping:

- Assign **Server** a fixed static address configuration matching the network block (e.g., **192.168.20.1**).
- Enable the **SMTP** and **POP3** capabilities inside the global services settings to accommodate active mail alerting pipelines.
- Access the central web manager platform to establish conditional **Action Rules**:
 - **Condition:** **If** Motion Detector status reads true (**Motion Detected? = Yes**).
 - **Actions:** **Then** force Siren and Light states to active (**ON**), initiate continuous Webcam video loops, and trigger an automated notification script to distribute an emergency status mail to the destination inbox.

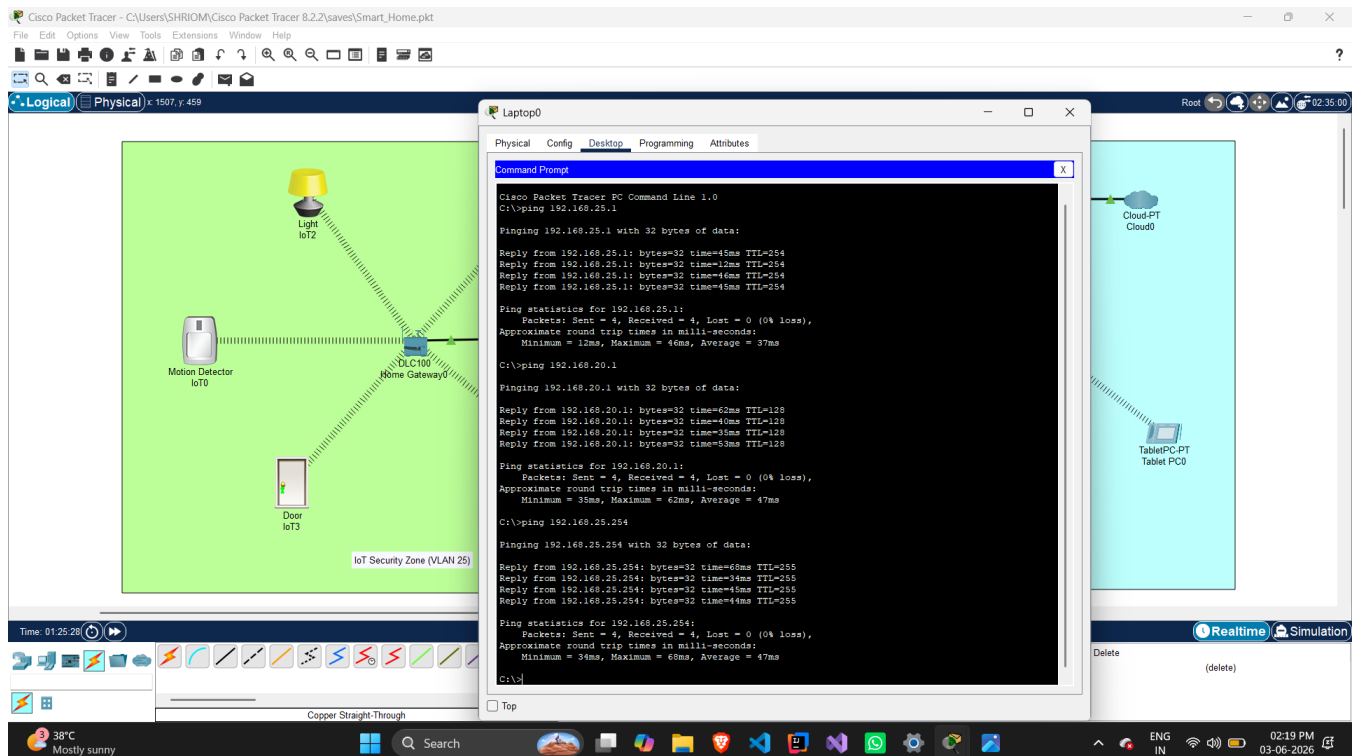
3. Testing and Verification

To validate the successful implementation and core functionality of the Smart Home Security Ecosystem, a rigorous three-phase verification process was conducted.

Step 3.1 Network Layer Verification (Inter-VLAN Routing)

To ensure that security boundaries do not obstruct cross-network remote access, network layer connectivity between the End-User Access Zone (VLAN 20) and the IoT Security Zone (VLAN 25) was verified.

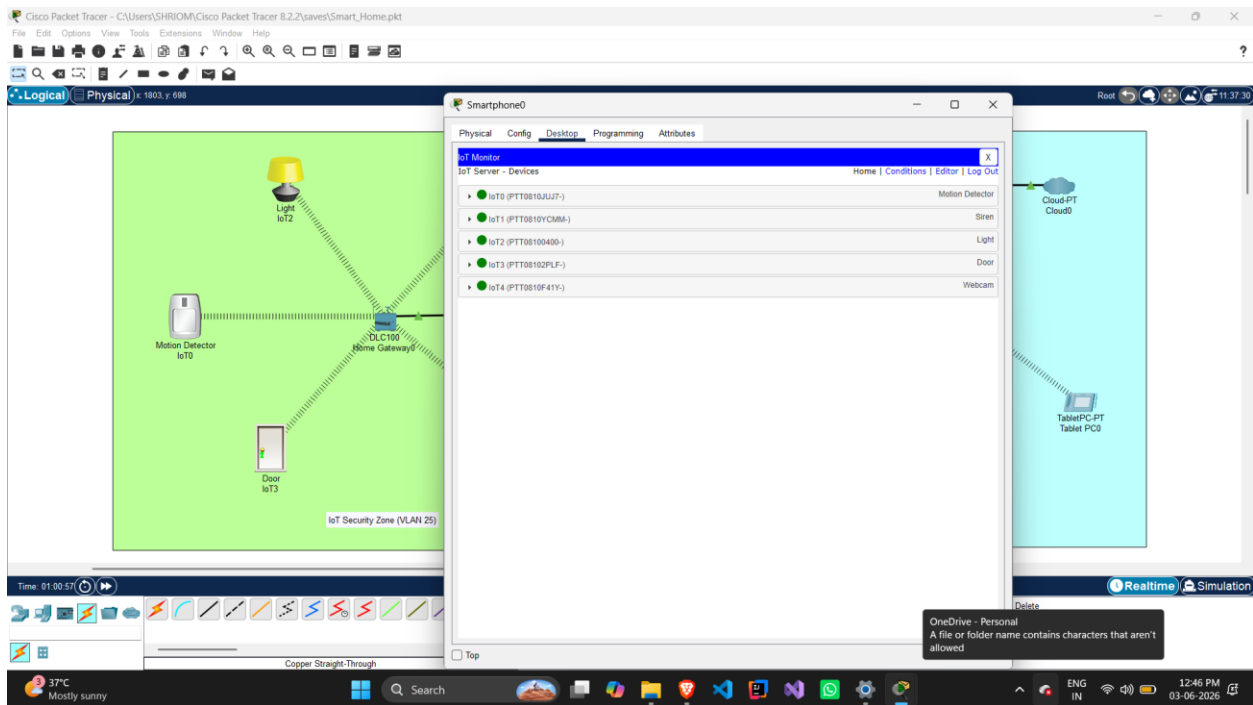
- **Command Prompt Ping Test:** A network ping was initiated from a workstation terminal (*PC0/Laptop0*) in VLAN 20 directly to the localized interface of *Home Gateway0* (192.168.25.1) in VLAN 25.
- **Result:** The ICMP packets successfully traversed the sub-interfaces (*Fa0/1.20* to *Fa0/1.25*) of *Router* with a 100% success rate, confirming stable Inter-VLAN deployment.



Step 3.2 Application Layer Verification (IoT Monitor Access)

This phase tests whether endpoint devices can securely interface with the smart home architecture to oversee live hardware parameters.

- **Remote Dashboard Verification:** A web session was initiated from the wireless **Tablet0** / **Smartphone0** by navigating to the gateway terminal address **http://192.168.25.1** and applying administrative validation tokens (**admin/admin**).
- **Result:** The interface completely synchronized, reflecting a live list of all localized smart peripherals (Motion Detector, Smart Light, Siren, Automated Door, Webcam) and proving active multi-network orchestration.



Step 3.3 IoT Automation Logic & Condition Settings

To enable autonomous threat response without manual user intervention, conditional logic rules were configured within the **Home Gateway** ecosystem. The following event-trigger framework was deployed and verified:

Rule 1: Intrusion Detection & Alarm Trigger

- **Logic Condition:** If the **Motion Detector** state is changed to **True** (Motion Detected).

- **Action Pipeline:**

- Set **Siren** status to **On** (Audio-visual alert activated).
- Set **Smart_Light** status to **On / Dim/Bright** (Illuminates the intrusion zone).
- Set **Door** status to **Lock**
- Set **Webcam** status to **On** (Initiates live video surveillance recording).

Rule 2: Automated Environment Reset (No Motion)

- **Logic Condition:** If the **Motion Detector** state returns to **False** (No Motion for a prolonged duration).

- **Action Pipeline:**

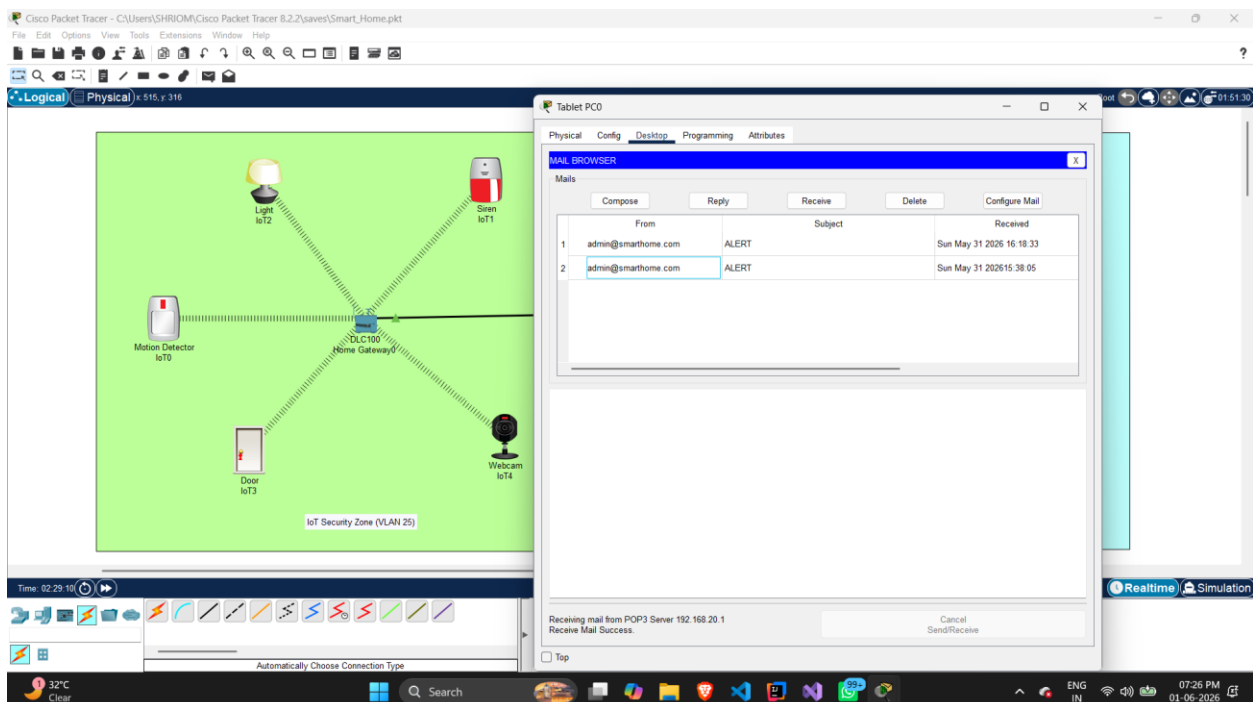
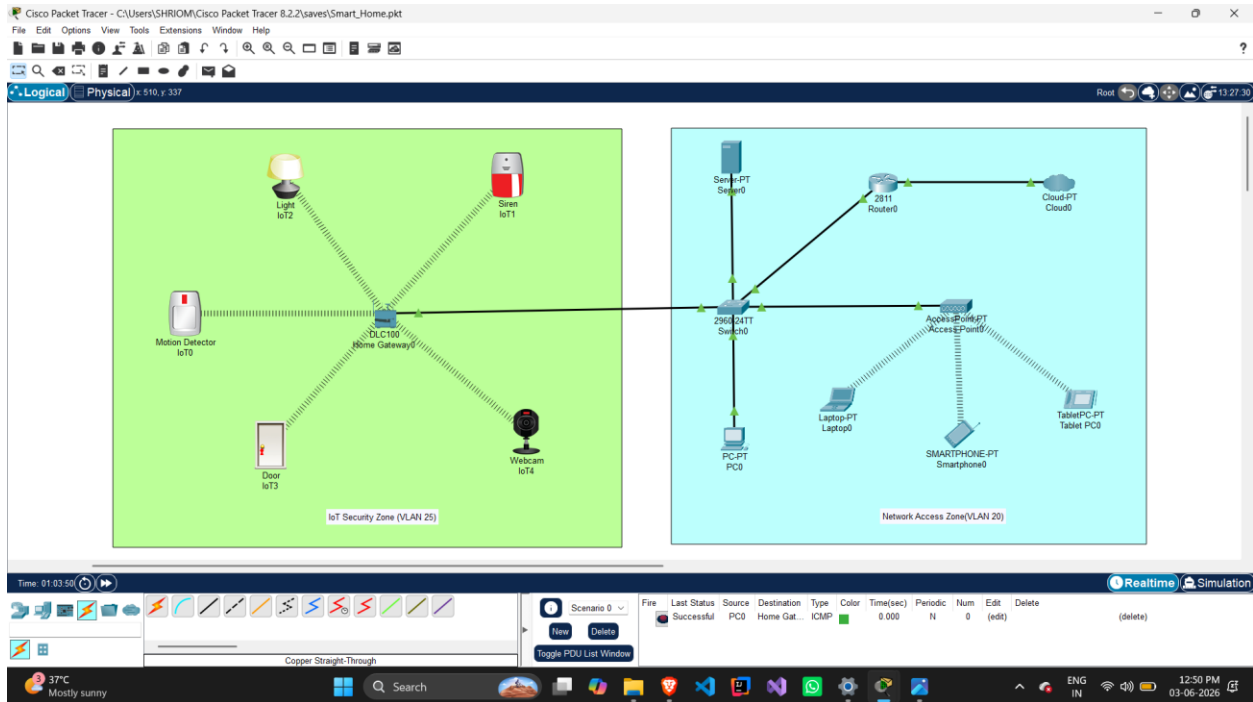
- Set **Siren** status to **Off** (Mutes the alarm).
- Set **Smart_Light** status to **Off** (Power-saving mode).
- Set **Door** status to **Unlock**
- Set **Webcam** status to **Off** (Standby mode).



Step 3.4 Security Logic & Automation Verification (Event-Trigger Loop)

The final phase tests the integrity of the conditional event pipeline inside the environment.

- **Threat Simulation:** Environmental movement was simulated on the **Motion Detector**.
- **Result:** The system triggered immediately. The local ecosystem responded instantly by pulsing the **Siren to ON**, illuminating the **Smart Light**, and initiating the **Webcam** loop. Concurrently, an email alert payload was compiled and routed via **Server** (SMTP) to the destination terminal inbox.



Conclusion

The design and successful implementation of the **Smart Home Security Ecosystem** using Cisco Packet Tracer effectively demonstrates the integration of advanced networking and the Internet of Things (IoT). By dividing the architecture into separate broadcast domains through **VLAN Segmentation** (VLAN 20 for user devices and VLAN 25 for critical IoT peripherals), the project ensures enhanced network performance, reduced broadcast traffic, and robust security limits.

The core challenge of enabling remote network management while preserving strict security boundaries was solved through the configuration of **Inter-VLAN Routing (Router-on-a-Stick)**. This allowed end-user terminals (like laptops and wireless tablets) to successfully communicate across network boundaries, monitor live metrics, and securely access the central management dashboard without compromising the privacy of the isolated IoT infrastructure.

The system was configured to detect motion events, activate security devices, and generate email-based alert notifications to users. The project successfully demonstrates the practical application of IoT, VLAN technology, network security, and automation in a smart home environment.

Overall, the Smart Home Security Ecosystem provides a scalable, secure, and efficient solution for modern home security requirements while offering valuable hands-on experience in IoT networking and Cisco Packet Tracer simulation.